Signal-to-Shock

From headline to exposure: calibrated news risk signals for wholesale banking stress testing

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Problem and approach

The problem. A single headline can threaten a bank's book, but risk scores from text are rarely calibrated, and generic stress tests shock every asset the same way. Risk teams need to know which exposures an event actually touches, and how far to trust the signal.

The approach.

1

Extract: an LLM with a FinBERT cross-check turns news and social posts into sentiment, event class, impact (1 to 10) and confidence.

2

Calibrate: map impact scores to abnormal returns observed in 24 historical market events.

3

Stress: trigger only confident, high-impact events, and shock only the exposures they touch.

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System design

1

Ingest and cache: GDELT, NewsAPI and social posts; every external call cached

2

Extract: LLM structured JSON with a FinBERT sentiment cross-check

3

Confidence and calibration: five-factor confidence; impact mapped to historical abnormal returns

4

Trigger gate: impact above 7.0, confidence at least 0.60, 48-hour debounce

5

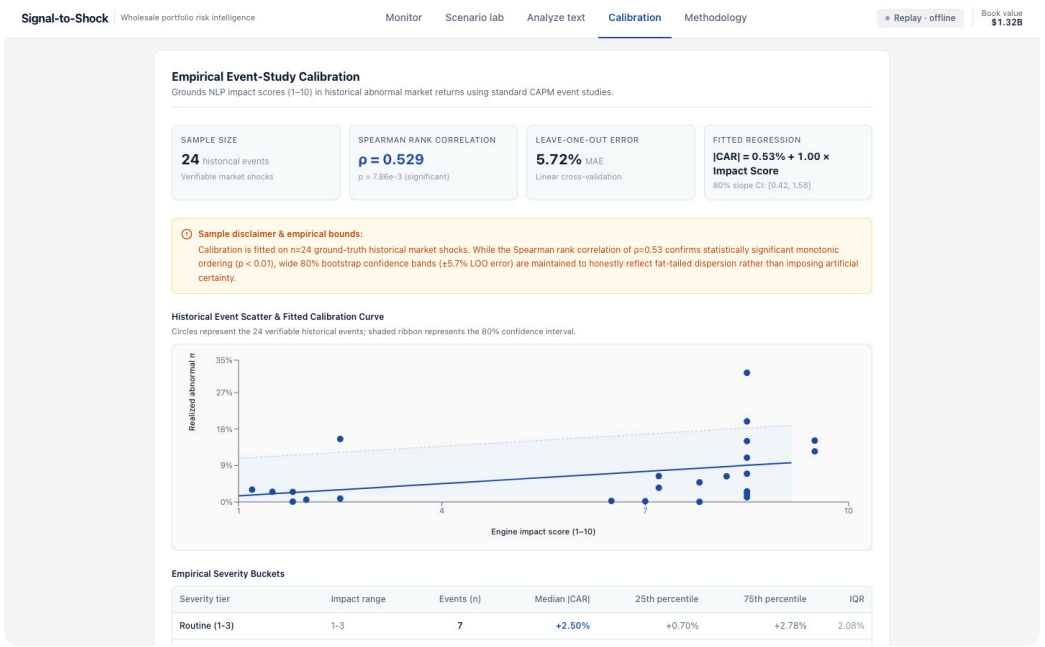
Stress and dashboard: theme-by-exposure shock table applied to a synthetic $1.32B book

Offline-first: every external call is cached, and a deterministic replay runs without keys or network.

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Implementation highlights

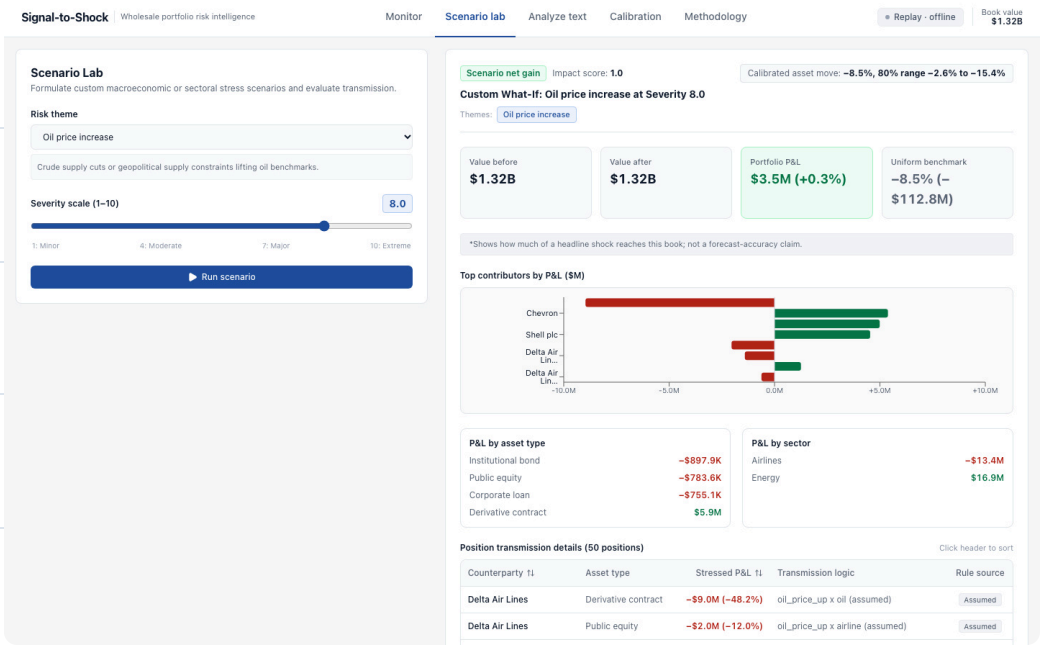
- **Structured extraction:** strict JSON, validated schema, controlled vocabulary of risk themes
- **Independent cross-check:** FinBERT sentiment agreement feeds the confidence score
- **Confidence:** source reliability, cross-source corroboration, sentiment agreement, self-consistency and claim certainty; unverified rumors are capped at 0.42
- **Calibration:** 24 verified events, |CAR| = 0.53% + 1.00 × impact (%), with a bootstrap 80% band
- **Exposure-aware shocks:** 41 theme-by-exposure rules (7 measured from data, 34 documented assumptions) applied to a 50-position, $1.32B synthetic book
- **Stack:** Python, FastAPI, Pydantic, FinBERT, React



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Key results

Event	Impact	Exposure-aware P&L	Uniform-shock P&L
Red Sea shipping attack	8.8	−0.21%	−9.34%
Bank credit watch (downgrade risk)	8.2	−1.16%	−8.74%
Antitrust action on a chip maker	7.5	−0.73%	−8.04%
OPEC surprise production cut	8.0	+0.27%	−8.54%

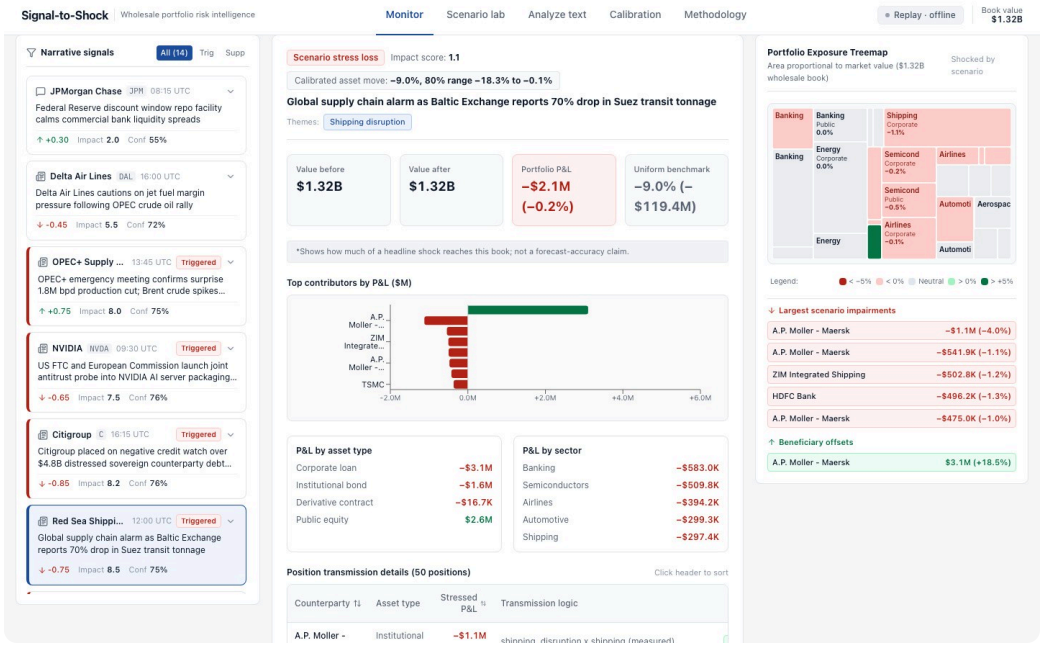


- 14 replayed signals: 5 passed the gate (merged into 4 scenarios); 9 were suppressed with stated reasons, including 3 social rumors
- Engine check on 25 labelled items: 76% event-class accuracy, impact error 1.26 points, rank correlation 0.68
- The uniform benchmark shows how much of a headline shock reaches this book. It is not a forecast-accuracy claim.

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Domain impact

- **Fewer false alarms:** confidence gating suppresses rumors and uncorroborated chatter
- **Explainable triggers:** every shock traces to a theme and exposure tag, labelled measured or assumed
- **Localised exposure:** the same event helps some positions and hurts others (an OPEC cut lifts energy producers and hits airlines)
- **Calibrated language:** "impact 8 historically meant roughly this move, within this range"
- **Reproducible and auditable:** offline replay, versioned data, open code



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Limitations and next steps

Limitations

- Calibration rests on 24 events; leave-one-out error is about 5.7%, so bands are wide
- 34 of 41 shock rules are assumptions, not measurements
- Portfolio and social posts are synthetic
- The LLM may recognise famous historical events
- Each signal maps to at most two themes; no second-order contagion yet

Next steps

- Larger event library and per-theme calibration
- Live streaming feeds with human review of triggers
- Contagion across counterparties and sectors
- Backtests against real loss data; threshold tuning with risk teams

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